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Passion for excellence in clean air solutions.

CASE STUDY NAME

FILTRATION OF ROBOTIC LASER WELDING IN AUTOMOBILE SUPPLY INDUSTRY



Case Study Facts

Product: PL-12000 Filter Unit	Process: Robotic laser welding in an automobile seating-systems production facility.

Application plant: TEKNİK MALZEME – FAURECIA Bursa Plant

teknik malzeme
a company of Faurecia and Orhan Holding

Requirements & Challenges

TEKNİK MALZEME is among the rare facilities in the world where automotive seats can be produced fully integrated. Its customers include the world's major automobile manufacturers such as Renault, Nissan, Volkswagen and Ford.

TEKNİK MALZEME required a filter unit / dust collector solution for its new invetsment about **robotic laser welding** in its seat production line. Due to the high welding speed and continuity in the new laser welding cabinets, where **ABB brand robots** are used, the filter unit (dust collector) had to extract the high amount of welding fume-smoke and dust load from the cabinets, so that it does not spread around in the production halls. Since the process has some fire risks, it was requested to integrate extra measures on the filter unit (dust collector) against possible fire risks.

SOLUTION

VANTERM took the data related to the robotic laser welding application from TEKNİK MALZEME and ABB, then carried out the necessary studies. VANTERM calculated the amount of extraction air required, depending on the planned laser welding process, material and capacities. It was decided to connect one filter unit (dust collector) to two welding cabinets, in order to save floor space in the production halls. Pressure losses of the duct system were calculated and analyzed, by VANTERM.

As a result **PL-12000 filter unit (dust collector)** is used in the prject. An automatic fire detection and extinguishing system was also integrated to the filter unit against the risk of fire..



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